

IET, DAVV, Indore
B.E. I year (IT-A+B)
Class Test – III (December -2023)
1AMRC1: Applied Mathematics-I

Time: 70 min.

Maximum Marks: 20

Note: Attempt only four questions.

- Q:1 If $u = \tan^{-1}(y^2/x)$, prove that $x^2 \partial^2 u / \partial x^2 + 2xy(\partial^2 u / \partial x \partial y) + y^2 \partial^2 u / \partial y^2 = -\sin^2 u \sin 2u$. (5)
- Q:2 Find the maxima and minima of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$. (5)
- Q:3 Find $\text{curl}(\text{curl } A)$ given, $A = x^2y \mathbf{i} + y^2z \mathbf{j} + z^2y \mathbf{k}$. (5)
- Q:4 Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = y^2 \mathbf{i} + xy \mathbf{j} + xz \mathbf{k}$ and C is the bounding curve of hemisphere $x^2 + y^2 + z^2 = 9, z > 0$, using stoke's theorem. (5)
- Q:5 Using Green's theorem, calculate $\oint_C [(3x + 2y) dx + (2y + 3x) dy]$, where $C: x^2 + y^2 = 1$. (5)